

Graphene and Graphene-Based Nano materials for Cancer Treatment

Review Article

Submitted by

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Organization	15+1
Introductions	15+1
Expt. Setup/ Analysis	15+1
Conclusion	15+1
Total	(60)+4 = (64)

1. INTRODUCTION

In this era of technological revolution, mankind has been rewarded with incredible innovations which has made life simpler and purposeful. From computers to self-driving cars using artificial intelligence, all of these have become an inseparable part of human life and science is the platform which helps all these innovations to evolve into something bigger every day. With the groundbreaking discovery of Graphene-The Wonder Material, technology has achieved a whole new level in recent times. Graphene's amazing potential is still being explored for various applications. From the time of its discovery, there have been numerous research encounters related to graphene. Edward, Dale et al. [1] have documented graphene's emergence as the basis of all technological evolutions over a decade after its discovery.

Researchers are exploring graphene's amazing potential every day for eclectic choices of fields ranging from electronics to medicine. Zhao, Hayner et al. used Si-Graphene composites for Lithium Ion batteries [2]. Further graphene has been used to develop energy efficient solar cells [3], desalination membranes [4]. Various researchers in the field of energy technology are also experimenting to incorporate graphene for producing cleaner fuel technologies involving biofuel cells, hydrogen storage and super-capacitors [5]-[7]. With numerous technological advancements, humankind still battles with the deadliest of diseases, like cancer.

1.1 Cancer

Cancer is the name given to a collection of related diseases. In all types of cancer, some of the body's cells begin to divide without stopping and spread into surrounding tissues. Cancer can start almost anywhere in the human body, which is made up of trillions of cells. Normally, human cells grow and divide to form new cells as the body needs them. When cells grow old or become damaged, they die, and new cells take their place. When cancer develops, however, this orderly process breaks down. As cells become more and more abnormal, old or damaged cells survive when they should die, and new cells form when they are not needed. These extra cells can divide without stopping and may form growths called tumors. Many cancers form solid tumors, which are masses of tissue. Cancers of the blood, such as leukemia, generally do not form solid tumors.

Cancerous tumors are malignant, which means they can spread into, or invade, nearby tissues. In addition, as these tumors grow, some cancer cells can break off and travel to distant places in the body through the blood or the lymph system and form new tumors far from the original tumor.

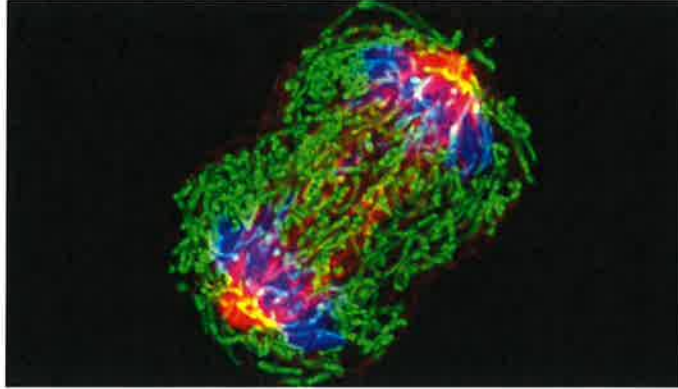


Figure 1. A dividing Breast Cancer Cell (Ref-National Cancer Institute/University of Pittsburgh Cancer Institute)

Unlike malignant tumors, benign tumors do not spread into, or invade, nearby tissues. Benign tumors can sometimes be quite large, however. When removed, they usually don't grow back, whereas malignant tumors sometimes do. Unlike most benign tumors elsewhere in the body, benign brain tumors can be life threatening. The process by which cancer cells spread to other parts of the body is called metastasis [8].

1.2 Types of cancers

Cancers are mostly named according to the organ in which it starts developing and also by the type of cell that formed them. There are several types of cancers, more than 100, but few of them occur in a majority.

Major types of cancer are;

- a. **Carcinoma**- These are formed by different types of epithelial cells and the nomenclature depends on the type of cell for example, Adenocarcinoma is formed by the cells that produce fluids or mucus. Basal Carcinoma develops in the lower layer of epidermis which leads to skin cancer. Many organs like stomach intestines, lungs, kidneys have squamous cell lining. These cells can also develop carcinoma cancer.

- b. **Sarcomas**- These are cancers that form in bone and soft tissues, including muscle, fat, blood vessels, lymph vessels, and fibrous tissue (such as tendons and ligaments).
- c. **Leukemia**- Cancers that begin in the blood-forming tissue of the bone marrow are called leukemia. These cancers do not form solid tumors. Instead, large numbers of abnormal white blood cells (leukemia cells and leukemic blast cells) build up in the blood and bone marrow, crowding out normal blood cells. The low level of normal blood cells can make it harder for the body to get oxygen to its tissues, control bleeding, or fight infections.
- d. **Lymphoma**- Lymphoma is cancer that begins in lymphocytes (T cells or B cells). These are disease-fighting white blood cells that are part of the immune system. In lymphoma, abnormal lymphocytes build up in lymph nodes and lymph vessels, as well as in other organs of the body.

1.3 Available Treatment Methods for Cancer

Treatment of cancer for anyone suffering with the disease can be an emotionally, mentally and physically enduring task. Most of the available methods of treatment alleviate patients of the suffering of cancer but these methods can sometimes prove to be ineffective for some of the cases and also comes with a lot of side-effects. Treatment method depends on the type of cancer and hence a combination of two-three methods can also be used for the whole procedure.

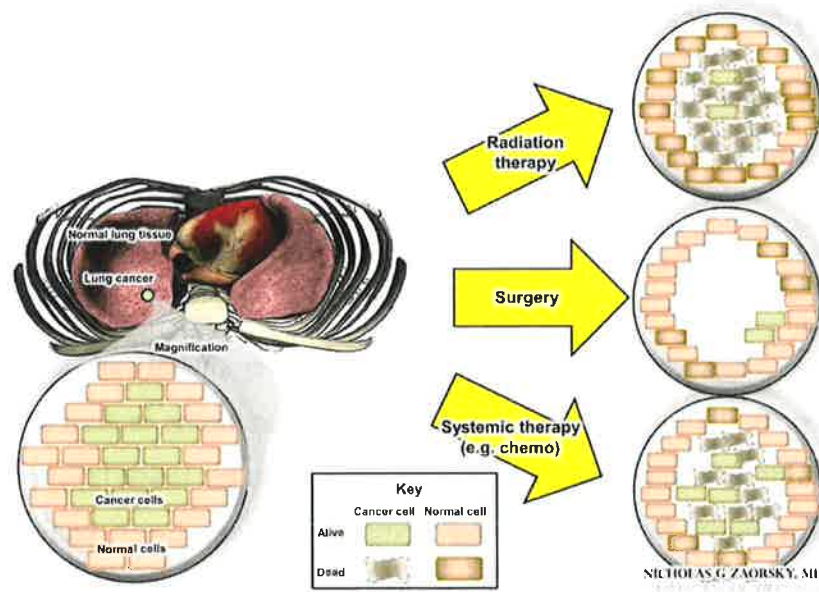


Figure 2. Common Types of Cancer Treatment Methods (Credit- CancerQuest)

The most common treatment methods available are;

- a. **Surgery**- In this process the surgeon removes the cancer from the affected organ by surgery.
- b. **Radiation Therapy**- This technique uses high levels of radiation to kill cancer cells and shrink tumors. This therapy has many side effects like destruction of healthy cells, fatigue, hair loss, blurry vision, etc.
- c. **Chemotherapy**- This method uses drugs to kill the cancer cells.
- d. **Immunotherapy**- This therapy helps the immune system to fight cancer cells.
- e. **Targeted Therapy**- It is a type of cancer treatment that targets the changes in cancer cells that help them grow, divide, and spread.
- f. **Hormone therapy**- It is a treatment that slows or stops the growth of breast and prostate cancers that use hormones to grow.
- g. **Stem cell transplants**- These are procedures that restore blood-forming stem cells in cancer patients who have had theirs destroyed by very high doses of chemotherapy or radiation therapy

1.4 Current Status of Cancer on a global level

Rebecca L. Seigel, Kimberley D. Miller, et al. present the current trends in cancer growth and the also propose the projected number of new cancer cases and deaths due to cancer in 2019. According to this study, 1,762,450 new cancer cases can arise with 606,880 deaths in 2019 in United States [9].

Estimated age-standardized incidence rates (World) in 2018, worldwide, both sexes, all ages

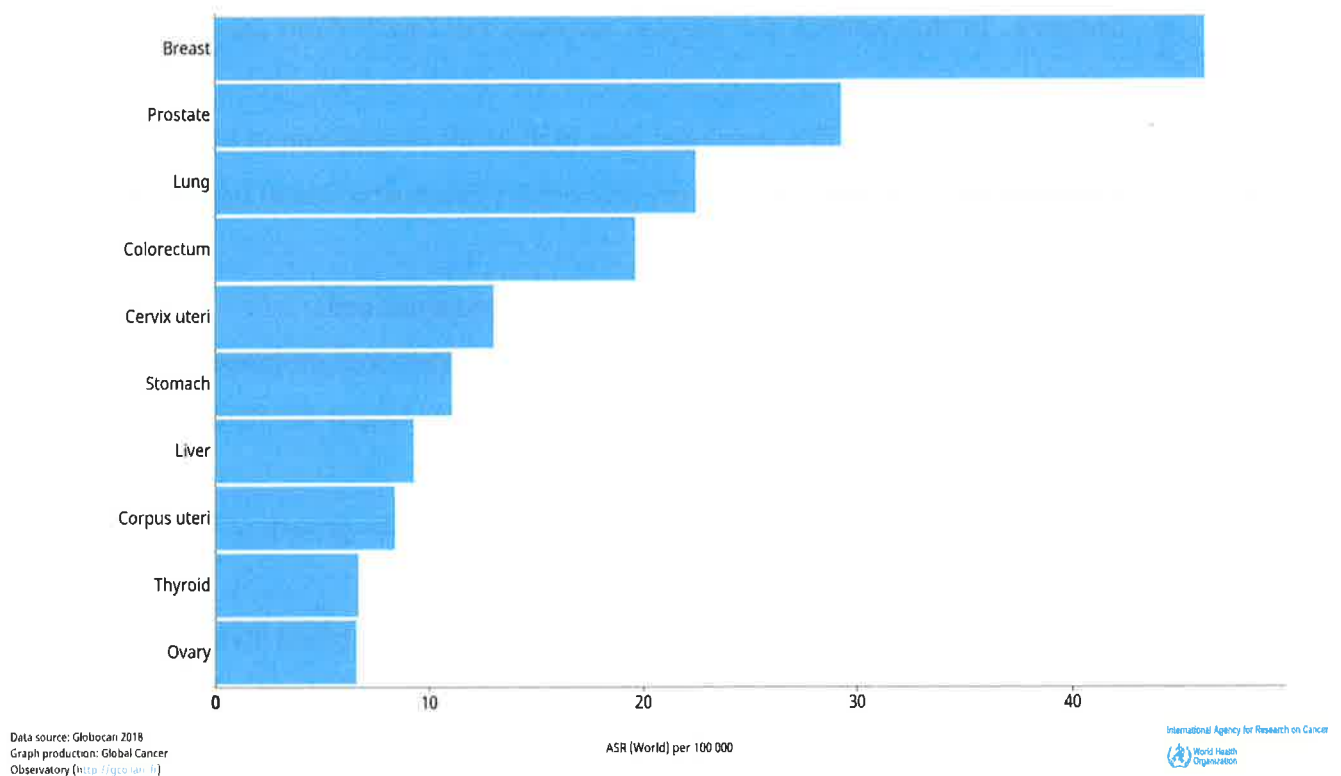


Figure 3. GLOBOCAN 2018 data for estimated cancer incidence rates worldwide

According to the Latest Global Cancer Data by International Agency for Research on Cancer over 9.6 million people died due to cancer in 2018 and 18.1 million new cases were reported [10]. With such an alarming death toll, serious implications and measures are required to get the situation under control. Cancer related awareness is a necessity for the people of the society including all age groups and genders. In the recent past, there have been advances in technology which has helped in early detection of cancer tumor cells and some of the treatment methods are proving lifesavers but a total cure for this deadly disease is still a difficult goal to achieve.

2. Graphene's role in cancer treatment

Graphene with extraordinary properties possesses the ability to be an important part in biomedical applications especially in cancer therapy. With a two dimensional shape and being the thinnest material in the world, graphene's strength and properties like high conductivity, high surface area, etc. proved to be the best for using it in the targeted drug delivery for cancer treatment. Many scientists and researchers have been studying graphene for its toxicity to the human body. Omid Akhvan et al. studied the genotoxicity of graphene nanoplatelets in human stem cells which showed that slight modifications in synthesized graphene can reduce toxic effects of graphene on human body [11].

Differentiation-Based Nano-Therapy

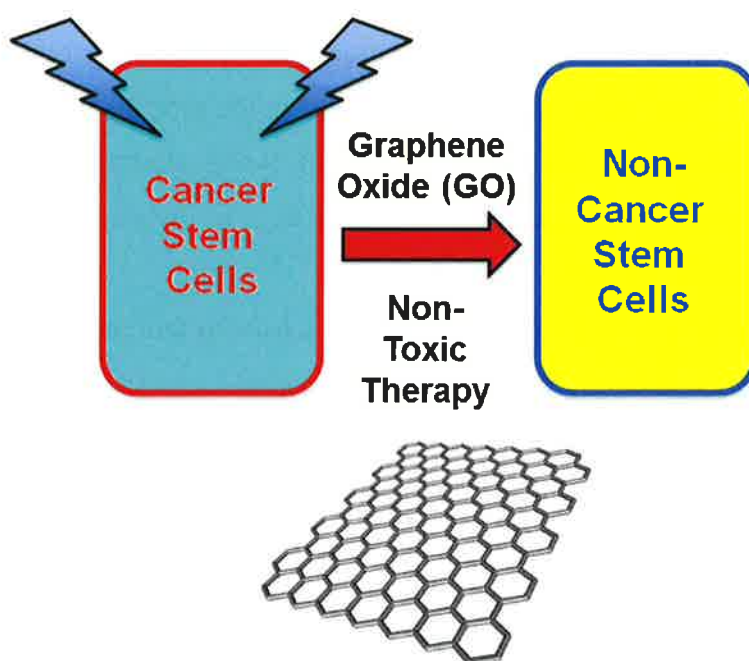


Figure 4. Schematic representation of use of graphene for cancer treatment

Maria Toloudi et al. find in their study that during the Targeted Therapeutic process the normal tissues are not affected, and this kind of treatment may be more effective and less cytotoxic than other non-targeted traditional chemotherapies [12]. Given the side effects and the destruction of healthy cells using other modes treatment, targeted drug delivery is an excellent option to reach

the particular cancer affected cells and destroy them using drugs leaving the surrounding cells healthy.

2.1 Graphene with Targeted delivery

Most types of targeted therapy help treat cancer by interfering with specific proteins that help tumors grow and spread throughout the body. They treat cancer in many different ways. They can:

- a. Help the immune system destroy cancer cells. One reason that cancer cells thrive is because they are able to hide from your immune system. Certain targeted therapies can mark cancer cells so it is easier for the immune system to find and destroy them. Other targeted therapies help boost your immune system to work better against cancer.
- b. Stop cancer cells from growing. Healthy cells in your body usually divide to make new cells only when they receive strong signals to do so. These signals bind to proteins on the cell surface, telling the cells to divide. This process helps new cells form only as your body needs them. But, some cancer cells have changes in the proteins on their surface that tell them to divide whether or not signals are present. Some targeted therapies interfere with these proteins, preventing them from telling the cells to divide. This process helps slow cancer's uncontrolled growth.
- c. Stop signals that help form blood vessels. Tumors need to form new blood vessels to grow beyond a certain size. In a process called angiogenesis, these new blood vessels form in response to signals from the tumor. Some targeted therapies called angiogenesis inhibitors are designed to interfere with these signals to prevent a blood supply from forming. Without a blood supply, tumors stay small. Or, if a tumor already has a blood supply, these treatments can cause blood vessels to die, which causes the tumor to shrink. Learn more about Angiogenesis Inhibitors.
- d. Deliver cell-killing substances to cancer cells. Some monoclonal antibodies are combined with toxins, chemotherapy drugs, and radiation. Once these monoclonal antibodies attach to targets on the surface of cancer cells, the cells take up the cell-killing substances, causing them to die. Cells that don't have the target will not be harmed.
- e. Cause cancer cell death. Healthy cells die in an orderly manner when they become damaged or are no longer needed. But, cancer cells have ways of avoiding this dying process. Some targeted therapies can cause cancer cells to go through this process of cell death.

- f. Starve cancer of the hormones it needs to grow. Some breast and prostate cancers require certain hormones to grow. Hormone therapies are a type of targeted therapy that can work in two ways. Some hormone therapies prevent your body from making specific hormones. Others prevent the hormones from acting on your cells, including cancer cells.

Smruthi Suryaprakash et al. used the combination of Mesenchymal stem cells (MSC) and Graphene Oxide (GO) to create an unique drug delivery platform wherein, GO can be used to load a range of therapeutics and MSC can carry the drug effectively to the site of the tumor. This could solve the problem of poor accumulation at the tumor site if GO alone is used.

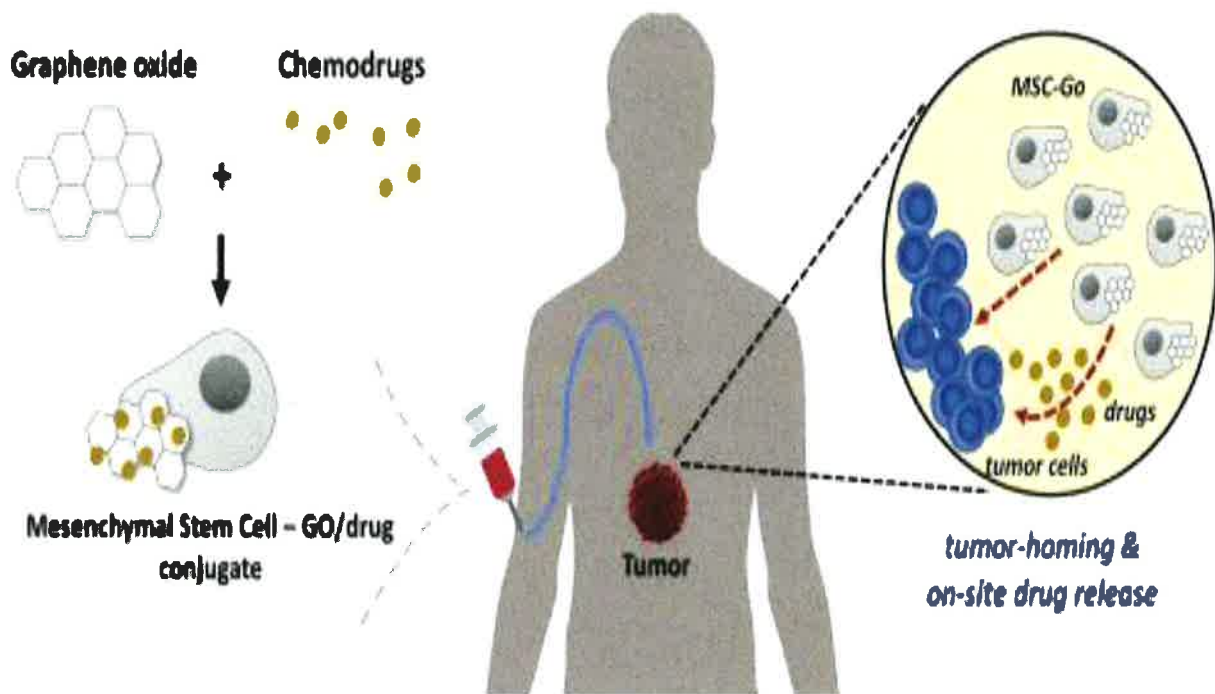


Figure 5. Combination of MSC and GO for effective cancer treatment.

3. Challenges and Conclusions

Although graphene with its potential as a wonder material is the best option for cancer treatment with targeted drug delivery methods, there are some challenges which still need to be resolved. These include the drawbacks of targeted drug delivery and also graphene's compatibility with the human body.

Targeted therapy have some drawbacks. These include:

- Cancer cells can become resistant to targeted therapy. For this reason, they may work best when used with other types of targeted therapy or with other cancer treatments, such as chemotherapy and radiation.
- Drugs for some targets are hard to develop. Reasons include the target's structure, the target's function in the cell, or both.

Schinwald et al. in their study found that graphene based nanoplatelets with aerodynamic diameter up to $25\mu\text{m}$ were respirable and inflammogenic in both the lung and the pleural space. Hence these are termed as novel hazards for the human health [14].

Graphene as every other material, comes with its pros and cons. But the extraordinary properties of graphene including both the physical and mechanical properties make it the most desirable material for most of the applications like electronics, biomedical and especially a significant one for cancer therapy.

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